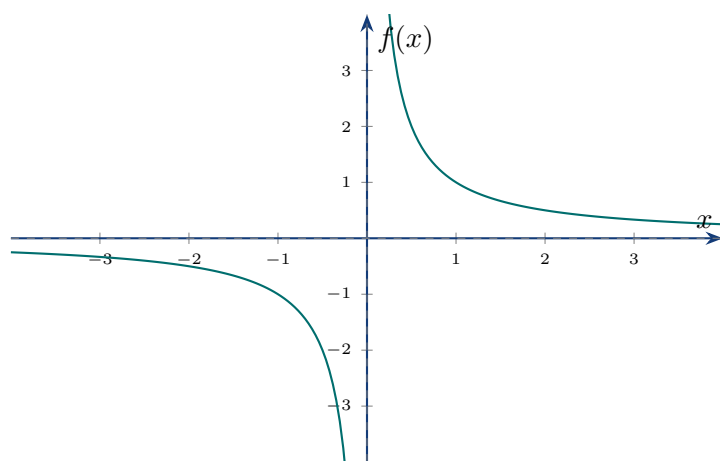


Limits & Continuity

A Complete Self-Contained Study Resource

for Semester Exam Preparation



Topics Covered:

Basic Concepts of Limits • One-Sided Limits • Infinite Limits

Limit Laws • Sandwich Theorem • Formal (ε - δ) Definitions

Continuity • Intermediate Value Theorem • Limits at Infinity & Asymptotes

Prepared for: **CSE 1-1 Students, Netrokona University**

Course: Calculus / Differential Calculus — Limits Chapter

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How to Use This Resource

This document is built to be read **once, start to finish**, and then used as a revision tool before the exam. Each topic follows the same structure:

1. **Intuition** – what is really going on, explained in plain language.
2. **Formal Definition / Notation** – the precise mathematical statement.
3. **Formula / Rule box** – the result you should memorize.
4. **Worked Examples** – fully solved, step by step.
5. **Practice Problems** – easy $\rightarrow$ medium $\rightarrow$ hard, for self-testing.

Colour-coded boxes are used throughout:

- **Blue** boxes = Definitions
- **Green** boxes = Formulas/Rules
- **Purple** boxes = Theorems
- **Yellow** boxes = Key notes / exam tips
- **Red** boxes = Common mistakes to avoid
- **Gray** boxes = Worked Examples

All practice-problem answers are collected in the **Answer Key** (final section) so you can self-check without spoilers while solving.

1 Basic Concepts of Limits

1.1 The Big Idea

Suppose you are walking towards a wall, and every step you take covers half the remaining distance. You never actually touch the wall in a finite number of steps, but the distance between you and the wall gets as small as you like. The wall's position is the **limit** of where you end up.

In calculus, a **limit** asks: “As x gets closer and closer to some number a (but is not necessarily equal to a), what value does $f(x)$ get closer and closer to?”

Key Note

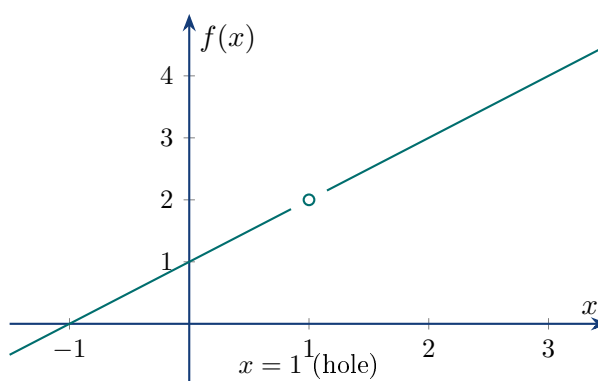
The limit of $f(x)$ as $x \rightarrow a$ describes the **behaviour of f near a** , not the value of f at a . The function does not even need to be defined at $x = a$ for the limit to exist.

1.2 A Motivating Table

Consider $f(x) = \frac{x^2 - 1}{x - 1}$. This function is undefined at $x = 1$ (division by zero), but we can still ask what happens as $x \rightarrow 1$.

x	0.9	0.99	0.999	$\rightarrow$	1	$\leftarrow$	1.001	1.01	1.1
$f(x)$	1.9	1.99	1.999	$\rightarrow$	2 (limit)	$\leftarrow$	2.001	2.01	2.1

Algebraically, $f(x) = \frac{(x-1)(x+1)}{x-1} = x+1$ for all $x \neq 1$. So the graph of f is just the line $y = x+1$ with a single point removed at $x = 1$ (a *hole*). Even though $f(1)$ is undefined, the limit as $x \rightarrow 1$ is clearly 2.



1.3 Informal Definition of a Limit

Definition

We say

$$\lim_{x \rightarrow a} f(x) = L$$

if the values of $f(x)$ can be made **arbitrarily close to L** by taking x **sufficiently close to a** (on either side), but not equal to a . We read this as “the limit of $f(x)$ as x approaches a equals L .”

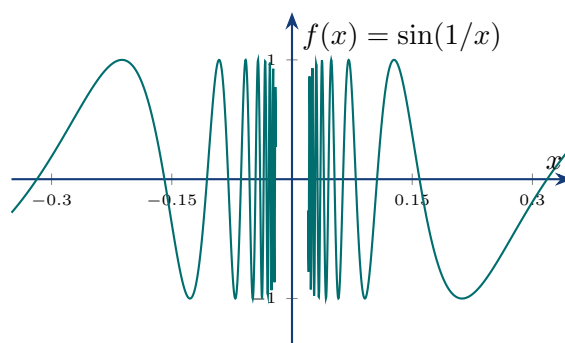
Key Note**Three important facts:**

- The limit only cares about values of x *near* a , never $x = a$ itself.
- $f(a)$ may or may not exist, and even if it exists, it need not equal L .
- A limit is a single number. If $f(x)$ does not settle down to one number, we say the limit **does not exist** (DNE).

1.4 When a Limit Fails to Exist

A limit fails to exist (at a finite value) typically for one of these reasons:

1. **Different left/right behaviour** – the function approaches different values from the left and right (jump discontinuity). Covered fully in Section 2.
2. **Unbounded growth** – the function grows without bound near a (vertical asymptote). Covered in Section 3.
3. **Oscillation** – the function oscillates infinitely often and never settles, e.g. $\sin(1/x)$ as $x \rightarrow 0$.

**1.5 Worked Examples****Example 1.1: Estimating a limit numerically**

Estimate $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$ using a table of values.

Solution. Factor: $\frac{x^2 - 4}{x - 2} = \frac{(x - 2)(x + 2)}{x - 2} = x + 2$ for $x \neq 2$.

As $x \rightarrow 2$ from either side, $x + 2 \rightarrow 4$. Checking with values $x = 1.9, 1.99, 2.01, 2.1$ gives $f(x) = 3.9, 3.99, 4.01, 4.1$ – clearly approaching 4.

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = 4.$$

Example 1.2: A limit that does not exist due to oscillation

Explain why $\lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right)$ does not exist.

Solution. As $x \rightarrow 0$, $\frac{1}{x} \rightarrow \pm\infty$, so the angle inside sine sweeps through infinitely many full cycles. No matter how small an interval around $x = 0$ we choose, $\sin(1/x)$ still takes *every* value between -1 and 1 inside that interval. It never settles near a single number L , so the limit does not exist.

Example 1.3: Limit exists even though function is undefined at the point

Find $\lim_{x \rightarrow 0} \frac{\sin x}{x}$, given the table below.

x (rad)	± 0.1	± 0.01	± 0.001
$\sin x/x$	0.99833	0.999983	0.99999983

Solution. The function is undefined at $x = 0$ ($0/0$ form), but the table shows $\frac{\sin x}{x} \rightarrow 1$ from both sides.

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1.$$

This is one of the most important standard limits in calculus – memorize it.

1.6 Practice Problems – Section 1**Easy**

- Using a table of values, estimate $\lim_{x \rightarrow 3} (2x + 1)$.
- Evaluate $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$ by factoring.
- Is it possible for $\lim_{x \rightarrow a} f(x)$ to exist when $f(a)$ is undefined? Justify briefly.

Medium

- Evaluate $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2}$.
- Sketch (by describing) the graph of a function f with $f(2) = 5$ but $\lim_{x \rightarrow 2} f(x) = 3$.
- Evaluate $\lim_{x \rightarrow 4} \frac{\sqrt{x} - 2}{x - 4}$ by rationalizing the numerator.

Hard

- Explain, with reasoning, why $\lim_{x \rightarrow 0} \frac{1}{x} \sin\left(\frac{1}{x}\right)$ does not exist even though $\sin(1/x)$ is bounded.
- Construct a piecewise function $g(x)$ such that $\lim_{x \rightarrow 0} g(x)$ exists and equals 2, but g is undefined at infinitely many points arbitrarily close to 0.

2 One-Sided Limits

2.1 The Big Idea

Sometimes a function behaves completely differently depending on whether x approaches a from the **left** (values less than a) or from the **right** (values greater than a). A classic real-life analogy: a shop's price for a product might jump at a certain quantity (e.g., “buy 10 or more, price per unit drops”). Just below 10 units the per-unit price is one value; at and above 10 it is another. One-sided limits let us describe such jump behaviour precisely.

2.2 Definitions

Left-hand and Right-hand Limits

Right-hand limit: $\lim_{x \rightarrow a^+} f(x) = L$ means $f(x) \rightarrow L$ as x approaches a through values *greater than* a (from the right).

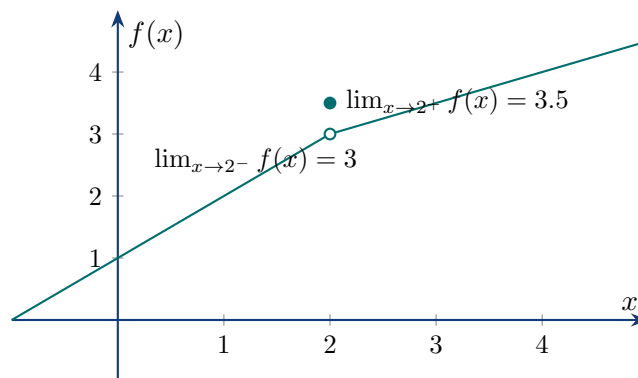
Left-hand limit: $\lim_{x \rightarrow a^-} f(x) = M$ means $f(x) \rightarrow M$ as x approaches a through values *less than* a (from the left).

Existence Rule Connecting One-Sided and Two-Sided Limits

$$\lim_{x \rightarrow a} f(x) = L \iff \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = L.$$

The two-sided limit exists **if and only if** both one-sided limits exist and are equal. If they differ, the two-sided limit does not exist (DNE).

2.3 Graphical Illustration – Jump Discontinuity



Here the left-hand limit (3) and right-hand limit (3.5) at $x = 2$ are different, so $\lim_{x \rightarrow 2} f(x)$ **does not exist**, even though the function is perfectly well-behaved on each side individually.

Key Note

Where one-sided limits are essential:

- Piecewise-defined functions, at the “joint” points.
- Functions defined only on one side of a point, e.g. $\sqrt{x}$ at $x = 0$ (only $\lim_{x \rightarrow 0^+}$ makes sense).
- The floor function $\lfloor x \rfloor$ and ceiling function $\lceil x \rceil$ at integers.
- Absolute value type expressions such as $\frac{|x|}{x}$.

2.4 Worked Examples

Example 2.1: Piecewise function

Let

$$f(x) = \begin{cases} x^2 + 1, & x < 1 \\ 3x - 1, & x \geq 1 \end{cases}$$

Find $\lim_{x \rightarrow 1^-} f(x)$, $\lim_{x \rightarrow 1^+} f(x)$, and determine whether $\lim_{x \rightarrow 1} f(x)$ exists.

Solution.

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (x^2 + 1) = 1^2 + 1 = 2.$$

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (3x - 1) = 3(1) - 1 = 2.$$

Both one-sided limits equal 2, so

$$\lim_{x \rightarrow 1} f(x) = 2.$$

(Note: this also happens to equal $f(1) = 3(1) - 1 = 2$, so f is actually continuous at $x = 1$ – see Section 6.)

Example 2.2: The absolute value trick

Evaluate $\lim_{x \rightarrow 0^+} \frac{|x|}{x}$ and $\lim_{x \rightarrow 0^-} \frac{|x|}{x}$.

Solution. For $x > 0$, $|x| = x$, so $\frac{|x|}{x} = 1$. Hence

$$\lim_{x \rightarrow 0^+} \frac{|x|}{x} = 1.$$

For $x < 0$, $|x| = -x$, so $\frac{|x|}{x} = \frac{-x}{x} = -1$. Hence

$$\lim_{x \rightarrow 0^-} \frac{|x|}{x} = -1.$$

Since the one-sided limits differ ($1 \neq -1$), $\lim_{x \rightarrow 0} \frac{|x|}{x}$ **does not exist**.

Example 2.3: Floor function

Evaluate $\lim_{x \rightarrow 3^-} \lfloor x \rfloor$ and $\lim_{x \rightarrow 3^+} \lfloor x \rfloor$, where $\lfloor x \rfloor$ is the greatest integer $\leq x$.

Solution. For x slightly less than 3 (e.g. 2.9, 2.99, ...), $\lfloor x \rfloor = 2$. So $\lim_{x \rightarrow 3^-} \lfloor x \rfloor = 2$.

For x slightly greater than or equal to 3 approaching from the right (e.g. 3.01, 3.1, ...), $\lfloor x \rfloor = 3$. So $\lim_{x \rightarrow 3^+} \lfloor x \rfloor = 3$.

Since $2 \neq 3$, $\lim_{x \rightarrow 3} \lfloor x \rfloor$ does not exist. This happens at **every integer** for the floor function.

Common Mistake

Students often confuse “the limit does not exist” with “the function is undefined.” A limit can fail to exist even when $f(a)$ is perfectly well-defined (Example 2.3: $\lfloor 3 \rfloor = 3$ is defined, but the two-sided limit still does not exist).

2.5 Practice Problems – Section 2**Easy**

1. For $f(x) = \begin{cases} x+2, & x < 0 \\ x^2, & x \geq 0 \end{cases}$, find $\lim_{x \rightarrow 0^-} f(x)$ and $\lim_{x \rightarrow 0^+} f(x)$.
2. Does $\lim_{x \rightarrow 0} \sqrt{x}$ exist as a two-sided limit? Explain using the domain of $\sqrt{x}$.
3. Evaluate $\lim_{x \rightarrow 5^+} \lfloor x \rfloor$ and $\lim_{x \rightarrow 5^-} \lfloor x \rfloor$.

Medium

4. For $g(x) = \begin{cases} 2x+1, & x < 2 \\ k, & x = 2 \\ x^2-3, & x > 2 \end{cases}$, find the value of k that would be needed for g to have a two-sided limit at $x = 2$ equal to that limit value (find the limit first).
5. Evaluate $\lim_{x \rightarrow 2^-} \frac{x-2}{|x-2|}$ and $\lim_{x \rightarrow 2^+} \frac{x-2}{|x-2|}$.
6. A function is defined as $h(x) = \frac{x^2-9}{x-3}$ for $x \neq 3$ and $h(3) = 10$. Find $\lim_{x \rightarrow 3^-} h(x)$ and $\lim_{x \rightarrow 3^+} h(x)$; state whether $\lim_{x \rightarrow 3} h(x) = h(3)$.

Hard

7. Construct a piecewise function with a removable discontinuity at $x = 1$ AND a jump discontinuity at $x = 3$, and compute all relevant one-sided limits to justify your construction.
8. Let $f(x) = \lfloor x \rfloor + \lfloor -x \rfloor$. Determine $\lim_{x \rightarrow n^-} f(x)$ and $\lim_{x \rightarrow n^+} f(x)$ for integer n , and hence describe f 's behaviour at integers.

3 Infinite Limits

3.1 The Big Idea

Sometimes as x approaches a point a , the function values do not settle near any finite number – instead they grow without bound (shoot up to $+\infty$) or shrink without bound (plunge to $-\infty$). This is called an **infinite limit**, and it signals a **vertical asymptote** at $x = a$.

Key Note

Writing $\lim_{x \rightarrow a} f(x) = \infty$ does **not** mean the limit exists. ∞ is not a number – this notation is a shorthand describing *why* the limit fails to exist (unbounded growth), not a value the limit equals.

3.2 Formal-Style Definition

Definition

We say $\lim_{x \rightarrow a} f(x) = \infty$ if for every $M > 0$, there exists $\delta > 0$ such that

$$0 < |x - a| < \delta \implies f(x) > M.$$

In words: $f(x)$ can be made **larger than any chosen bound** M by taking x close enough to a . Similarly, $\lim_{x \rightarrow a} f(x) = -\infty$ if $f(x)$ can be made **smaller than any chosen (very negative) bound** by taking x close enough to a .

3.3 Vertical Asymptotes

Vertical Asymptote

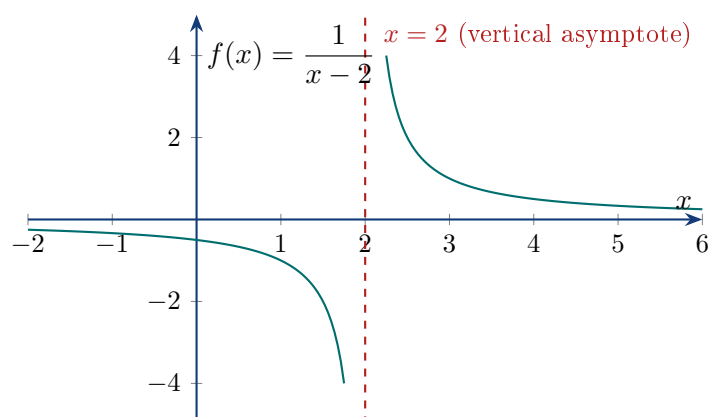
The line $x = a$ is a **vertical asymptote** of $y = f(x)$ if at least one of the following holds:

$$\lim_{x \rightarrow a^-} f(x) = \pm\infty, \quad \text{or} \quad \lim_{x \rightarrow a^+} f(x) = \pm\infty.$$

Quick Test for Rational Functions

For $f(x) = \frac{p(x)}{q(x)}$ in lowest terms, $x = a$ is a vertical asymptote whenever $q(a) = 0$ but $p(a) \neq 0$. Determine the sign of $f(x)$ just to the left and right of a by checking the sign of numerator and denominator there – this tells you whether the branch goes to $+\infty$ or $-\infty$.

3.4 Graphical Illustration



Here: $\lim_{x \rightarrow 2^-} \frac{1}{x-2} = -\infty$ (denominator is a small negative number) and $\lim_{x \rightarrow 2^+} \frac{1}{x-2} = +\infty$ (denominator is a small positive number).

3.5 Worked Examples

Example 3.1: Basic sign analysis

Evaluate $\lim_{x \rightarrow 3^+} \frac{2}{x-3}$ and $\lim_{x \rightarrow 3^-} \frac{2}{x-3}$.

Solution. As $x \rightarrow 3^+$: $x-3$ is a small **positive** number, so $\frac{2}{x-3}$ is a large **positive** number.

$$\lim_{x \rightarrow 3^+} \frac{2}{x-3} = +\infty.$$

As $x \rightarrow 3^-$: $x-3$ is a small **negative** number, so $\frac{2}{x-3}$ is a large **negative** number.

$$\lim_{x \rightarrow 3^-} \frac{2}{x-3} = -\infty.$$

Since the one-sided infinite behaviours differ in sign, $\lim_{x \rightarrow 3} \frac{2}{x-3}$ does not exist (not even as $\pm\infty$), but $x=3$ is still a vertical asymptote.

Example 3.2: Even power denominator

Evaluate $\lim_{x \rightarrow 1} \frac{-3}{(x-1)^2}$.

Solution. Since $(x-1)^2 > 0$ for all $x \neq 1$, the denominator is always positive near $x=1$ (approaching 0^+ from both sides). The numerator is -3 (negative constant).

negative $\div$ small positive = large negative.

This holds from *both* sides, so

$$\lim_{x \rightarrow 1^-} \frac{-3}{(x-1)^2} = \lim_{x \rightarrow 1^+} \frac{-3}{(x-1)^2} = -\infty \implies \lim_{x \rightarrow 1} \frac{-3}{(x-1)^2} = -\infty.$$

(We can write this because both one-sided “limits” agree on the same infinite behaviour.)

Example 3.3: Rational function with factoring

Find all vertical asymptotes of $f(x) = \frac{x+1}{x^2-4}$ and describe the sign behaviour near each.

Solution. Factor the denominator: $x^2-4 = (x-2)(x+2)$. Zeros at $x=2$ and $x=-2$; numerator $x+1 \neq 0$ at either point, so both are vertical asymptotes.

At $x=2$: numerator $\rightarrow 3$ (positive). Just left of 2 ($x=1.9$): $(x-2) < 0$, $(x+2) > 0 \Rightarrow$ denom < 0 , so $f \rightarrow -\infty$. Just right of 2 ($x=2.1$): denom > 0 , so $f \rightarrow +\infty$.

At $x=-2$: numerator $\rightarrow -1$ (negative). Just left of -2 ($x=-2.1$): $(x-2) < 0$, $(x+2) < 0 \Rightarrow$ denom > 0 , so $f \rightarrow -\infty$ (sign check), giving $f \rightarrow$ negative/positive $= -\infty$. Just right of -2 ($x=-1.9$): $(x-2) < 0$, $(x+2) > 0 \Rightarrow$ denom < 0 , so $f \rightarrow$ negative/negative $= +\infty$.

Common Mistake

Do not simply cancel factors carelessly. If *both* numerator and denominator vanish at $x = a$, that is a **removable discontinuity** (a hole), **not** a vertical asymptote – see the difference between Example 1.1 (hole) and this section (asymptote). Always check whether the numerator is also zero at the candidate point.

3.6 Practice Problems – Section 3**Easy**

1. Evaluate $\lim_{x \rightarrow 0^+} \frac{1}{x}$ and $\lim_{x \rightarrow 0^-} \frac{1}{x}$.
2. Identify the vertical asymptote(s) of $f(x) = \frac{5}{x+4}$.
3. Evaluate $\lim_{x \rightarrow 0} \frac{4}{x^2}$.

Medium

4. Find all vertical asymptotes of $f(x) = \frac{x-1}{x^2-x-6}$ and classify the sign behaviour on each side.
5. Evaluate $\lim_{x \rightarrow 2^+} \frac{x}{x-2}$ and $\lim_{x \rightarrow 2^-} \frac{x}{x-2}$.
6. Determine whether $f(x) = \frac{x^2-9}{x-3}$ has a vertical asymptote at $x = 3$, and justify using the numerator/denominator zero test.

Hard

7. Find the vertical asymptotes of $f(x) = \frac{x^2-4}{x^3-3x^2-4x}$ (fully factor first), and determine one-sided infinite limits at each.
8. For $f(x) = \frac{1}{x-a} + \frac{1}{x-b}$ ($a \neq b$), describe how many vertical asymptotes exist and the sign behaviour near each, in terms of a and b .

4 Limit Laws

4.1 The Big Idea

Computing limits from a table or graph every time is slow. Once we know a few "safe" building-block limits and a set of rules for combining them (sums, products, quotients...), we can compute almost any limit **algebraically**, quickly and exactly. These rules are the **Limit Laws**.

Key Note

All laws below assume $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$ both **exist** (are finite real numbers).

4.2 The Laws

Limit Laws (assuming the individual limits exist)

1. **Constant:** $\lim_{x \rightarrow a} c = c$
2. **Identity:** $\lim_{x \rightarrow a} x = a$
3. **Sum/Difference:** $\lim_{x \rightarrow a} [f(x) \pm g(x)] = L \pm M$
4. **Constant Multiple:** $\lim_{x \rightarrow a} [c \cdot f(x)] = c \cdot L$
5. **Product:** $\lim_{x \rightarrow a} [f(x) \cdot g(x)] = L \cdot M$
6. **Quotient:** $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{L}{M}$, provided $M \neq 0$
7. **Power:** $\lim_{x \rightarrow a} [f(x)]^n = L^n$, $n \in \mathbb{Z}^+$
8. **Root:** $\lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{L}$, (with $L \geq 0$ if n even)

Direct Substitution Property

If f is a polynomial, or a rational function with a nonzero denominator at a , or built from standard functions (trig, exponential, log, root) at a point in their domain, then

$$\lim_{x \rightarrow a} f(x) = f(a).$$

This is the fastest way to evaluate a limit – **try direct substitution first**. Only fall back to algebraic tricks (factoring, rationalizing) when substitution gives an **indeterminate form** like $\frac{0}{0}$.

4.3 Strategy for Indeterminate Forms

When direct substitution gives $\frac{0}{0}$ (an **indeterminate form**), the limit may still exist – you must simplify first. Standard techniques:

1. **Factor and cancel** common factors (Example 4.1).
2. **Rationalize** the numerator or denominator when square roots are present (Example 4.2).
3. **Simplify complex fractions** by combining into a single fraction first (Example 4.3).
4. **Use standard trigonometric limits:** $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$, $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$.

Common Mistake

$\frac{0}{0}$ and $\frac{\infty}{\infty}$ are **indeterminate**, meaning they do *not* tell you the answer – the limit could be any real number, $\pm\infty$, or DNE. You **must** simplify algebraically before concluding anything. Never write “ $= \frac{0}{0}$ ” as a final answer.

4.4 Worked Examples**Example 4.1: Factor and cancel**

Evaluate $\lim_{x \rightarrow -1} \frac{x^2 + 3x + 2}{x + 1}$.

Solution. Direct substitution: $\frac{1 - 3 + 2}{0} = \frac{0}{0}$ – indeterminate, so factor.

$$\frac{x^2 + 3x + 2}{x + 1} = \frac{(x + 1)(x + 2)}{x + 1} = x + 2, \quad (x \neq -1).$$

$$\lim_{x \rightarrow -1} \frac{x^2 + 3x + 2}{x + 1} = \lim_{x \rightarrow -1} (x + 2) = 1.$$

Example 4.2: Rationalizing

Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{x}$.

Solution. Direct substitution gives $\frac{0}{0}$. Multiply numerator and denominator by the conjugate $\sqrt{x+4} + 2$:

$$\frac{\sqrt{x+4} - 2}{x} \cdot \frac{\sqrt{x+4} + 2}{\sqrt{x+4} + 2} = \frac{(x+4) - 4}{x(\sqrt{x+4} + 2)} = \frac{x}{x(\sqrt{x+4} + 2)} = \frac{1}{\sqrt{x+4} + 2}, \quad (x \neq 0).$$

Now substitute directly:

$$\lim_{x \rightarrow 0} \frac{1}{\sqrt{x+4} + 2} = \frac{1}{\sqrt{4} + 2} = \frac{1}{4}.$$

Example 4.3: Complex fraction

Evaluate $\lim_{x \rightarrow 2} \frac{\frac{1}{x} - \frac{1}{2}}{x - 2}$.

Solution. Combine the numerator into a single fraction:

$$\frac{1}{x} - \frac{1}{2} = \frac{2 - x}{2x}.$$

So the whole expression becomes

$$\frac{2 - x}{2x(x - 2)} = \frac{-(x - 2)}{2x(x - 2)} = \frac{-1}{2x}, \quad (x \neq 2).$$

$$\lim_{x \rightarrow 2} \frac{-1}{2x} = \frac{-1}{4}.$$

Example 4.4: Combining several laws

Given $\lim_{x \rightarrow 3} f(x) = 4$ and $\lim_{x \rightarrow 3} g(x) = -2$, evaluate $\lim_{x \rightarrow 3} \left[\frac{2f(x) - g(x)^2}{f(x) + 5} \right]$.

Solution. Apply the laws step by step (constant multiple, power, sum, quotient):

$$\lim_{x \rightarrow 3} 2f(x) = 2(4) = 8, \quad \lim_{x \rightarrow 3} g(x)^2 = (-2)^2 = 4.$$

$$\lim_{x \rightarrow 3} [2f(x) - g(x)^2] = 8 - 4 = 4.$$

$$\lim_{x \rightarrow 3} [f(x) + 5] = 4 + 5 = 9 \neq 0 \quad (\text{quotient law applies}).$$

$$\lim_{x \rightarrow 3} \left[\frac{2f(x) - g(x)^2}{f(x) + 5} \right] = \frac{4}{9}.$$

4.5 Practice Problems – Section 4**Easy**

1. Evaluate $\lim_{x \rightarrow 2} (3x^2 - 5x + 1)$ using direct substitution.
2. Evaluate $\lim_{x \rightarrow -3} \frac{x + 3}{x^2 - 9}$.
3. Given $\lim_{x \rightarrow 1} f(x) = 5$, $\lim_{x \rightarrow 1} g(x) = 2$, find $\lim_{x \rightarrow 1} [3f(x) - 2g(x)]$.

Medium

4. Evaluate $\lim_{x \rightarrow 4} \frac{\sqrt{x} - 2}{x - 4}$.
5. Evaluate $\lim_{x \rightarrow 0} \frac{(2 + x)^3 - 8}{x}$.
6. Evaluate $\lim_{h \rightarrow 0} \frac{(3 + h)^2 - 9}{h}$ (this is exactly the derivative definition pattern you will meet in the next chapter).

Hard

7. Evaluate $\lim_{x \rightarrow 1} \frac{x^3 - 1}{\sqrt{x} - 1}$ (hint: rationalize the denominator, then factor the numerator).
8. Evaluate $\lim_{x \rightarrow 0} \frac{1}{x} \left(\frac{1}{\sqrt{1+x}} - 1 \right)$.

5 The Sandwich (Squeeze) Theorem

5.1 The Big Idea

Imagine a function $g(x)$ is trapped between two other functions $f(x)$ and $h(x)$ near a point a , like a slice of cheese squeezed between two slices of bread. If both bread slices (top and bottom functions) come together at the *same* height as $x \rightarrow a$, the cheese in between is forced to go there too. This lets us find limits of complicated or oscillating functions by trapping them between two simple functions whose limits we already know.

5.2 Formal Statement

Sandwich (Squeeze) Theorem

Suppose that for all x in an open interval containing a (except possibly at a itself),

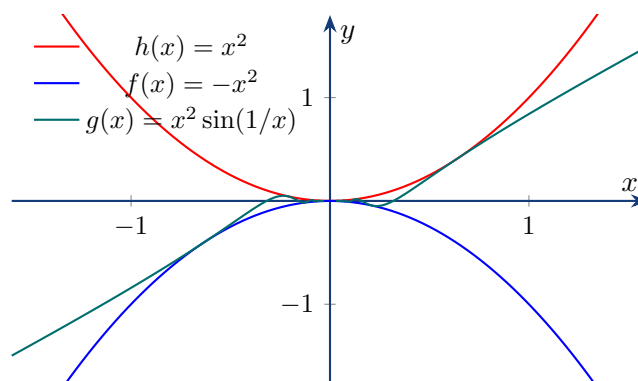
$$f(x) \leq g(x) \leq h(x),$$

and suppose

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L.$$

Then

$$\lim_{x \rightarrow a} g(x) = L \text{ as well.}$$



Both the red curve (x^2) and blue curve ($-x^2$) squeeze together to 0 at $x = 0$, so the teal oscillating curve trapped between them is **forced** to also approach 0.

Key Note

The Sandwich Theorem is the standard tool for:

- Limits involving sin or cos multiplied by something going to 0, e.g. $x^2 \sin(1/x)$.
- Proving the fundamental trigonometric limit $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ (using a geometric area argument with $\sin \theta \leq \theta \leq \tan \theta$).
- Bounding a hard-to-evaluate function between two easy ones.

The key skill is **finding the right bounding functions** f and h – almost always by using the fact that $-1 \leq \sin(\cdot) \leq 1$ and $-1 \leq \cos(\cdot) \leq 1$.

5.3 Worked Examples

Example 5.1: Classic oscillating example

Evaluate $\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right)$.

Solution. We cannot substitute directly ($\sin(1/x)$ is undefined at $x = 0$ and oscillates wildly). But we know for all $x \neq 0$:

$$-1 \leq \sin\left(\frac{1}{x}\right) \leq 1.$$

Multiply through by $x^2 \geq 0$ (multiplying by a nonnegative quantity does not flip inequalities):

$$-x^2 \leq x^2 \sin\left(\frac{1}{x}\right) \leq x^2.$$

Now, $\lim_{x \rightarrow 0} (-x^2) = 0$ and $\lim_{x \rightarrow 0} x^2 = 0$. Both bounding functions squeeze to 0, so by the Sandwich Theorem:

$$\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right) = 0.$$

Example 5.2: A variant with cosine

Evaluate $\lim_{x \rightarrow 0} x^4 \cos\left(\frac{2}{x^3}\right)$.

Solution. Since $-1 \leq \cos\left(\frac{2}{x^3}\right) \leq 1$ for all $x \neq 0$, and $x^4 \geq 0$:

$$-x^4 \leq x^4 \cos\left(\frac{2}{x^3}\right) \leq x^4.$$

As $x \rightarrow 0$, both $-x^4 \rightarrow 0$ and $x^4 \rightarrow 0$. By the Sandwich Theorem,

$$\lim_{x \rightarrow 0} x^4 \cos\left(\frac{2}{x^3}\right) = 0.$$

Example 5.3: Proving a basic sine limit using bounds

Show that $\lim_{x \rightarrow 0} \sin x = 0$ using the inequality $|\sin x| \leq |x|$ for all real x .

Solution. The inequality $|\sin x| \leq |x|$ gives $-|x| \leq \sin x \leq |x|$.

As $x \rightarrow 0$, both $-|x| \rightarrow 0$ and $|x| \rightarrow 0$ (direct substitution, since $|x|$ is continuous). By the Sandwich Theorem,

$$\lim_{x \rightarrow 0} \sin x = 0.$$

Common Mistake

The inequality you choose **must hold for all x near a** (not just at isolated points), and **both** bounding limits must be computable and equal. A common mistake is bounding with functions whose limits are different, or bounding only loosely so the two sides do not squeeze together.

5.4 Practice Problems – Section 5

Easy

1. Evaluate $\lim_{x \rightarrow 0} x^2 \cos\left(\frac{1}{x}\right)$ using the Sandwich Theorem.
2. Evaluate $\lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right)$.
3. If $2 - x^2 \leq g(x) \leq 2 + x^2$ for all x , find $\lim_{x \rightarrow 0} g(x)$.

Medium

4. Evaluate $\lim_{x \rightarrow 0} x^2 \sin^2\left(\frac{5}{x}\right)$.
5. If $1 - \frac{x^2}{6} \leq \frac{\sin x}{x} \leq 1$ for x near 0 (a known trig bound), what is $\lim_{x \rightarrow 0} \frac{\sin x}{x}$?
6. Evaluate $\lim_{x \rightarrow \infty} \frac{\sin x}{x}$ using an appropriate squeeze (careful: this is a limit at infinity, previewed here).

Hard

7. Evaluate $\lim_{x \rightarrow 0} x^2 e^{\sin(1/x)}$, given that $e^{-1} \leq e^{\sin(1/x)} \leq e$ for all $x \neq 0$.
8. Using the Sandwich Theorem and the known bound $\theta \cos \theta \leq \sin \theta \leq \theta$ for small positive θ , prove $\lim_{\theta \rightarrow 0^+} \frac{\sin \theta}{\theta} = 1$.

6 Formal Definition of Limits, and Continuity

6.1 The Big Idea Behind ε - δ

So far, “arbitrarily close” and “sufficiently close” have been informal phrases. The **formal** (ε - δ) **definition** turns these into precise, checkable mathematical statements. Think of it as a game:

- **Challenger** picks how close to L they want $f(x)$ to be – this tolerance is called ε (epsilon), always a positive number.
- **You** must respond with a “closeness to a ” – called δ (delta) – such that whenever x is within δ of a (but $x \neq a$), $f(x)$ is guaranteed to land within ε of L .

If you can *always* find a suitable δ , no matter how small ε is, then the limit truly equals L .

6.2 Formal Definition

ε - δ Definition of a Limit

We say

$$\lim_{x \rightarrow a} f(x) = L$$

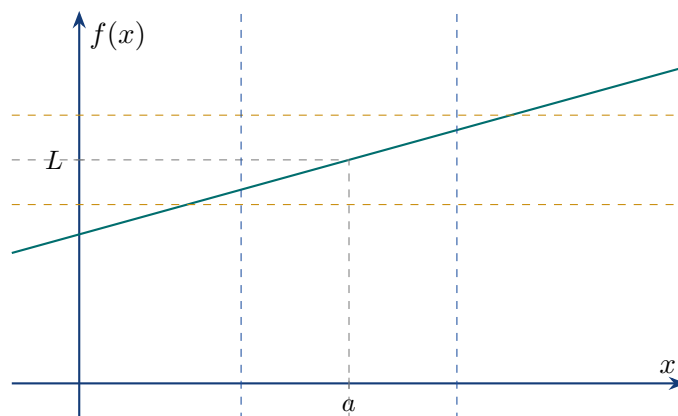
if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon.$$

Key Note

Reading the symbols: $|x - a| < \delta$ means “ x is within δ of a ”; $0 < |x - a|$ excludes $x = a$ itself; $|f(x) - L| < \varepsilon$ means “ $f(x)$ is within ε of L .” The definition says: *no matter how small a tolerance ε you demand around L , I can always find a small enough neighbourhood δ around a that guarantees it.*

6.3 Diagram



Whenever x lies in the vertical band $(a - \delta, a + \delta)$, the corresponding $f(x)$ is trapped inside the horizontal band $(L - \varepsilon, L + \varepsilon)$.

6.4 Worked Examples – Formal Proofs

Example 6.1: Proving a linear limit

Use the ε - δ definition to prove $\lim_{x \rightarrow 3} (2x - 1) = 5$.

Solution. *Scratch work:* we want $|(2x - 1) - 5| < \varepsilon$, i.e. $|2x - 6| < \varepsilon$, i.e. $2|x - 3| < \varepsilon$, i.e. $|x - 3| < \frac{\varepsilon}{2}$. This suggests choosing $\delta = \frac{\varepsilon}{2}$.

Formal proof: Let $\varepsilon > 0$ be given. Choose $\delta = \frac{\varepsilon}{2}$. Suppose $0 < |x - 3| < \delta$. Then

$$|(2x - 1) - 5| = |2x - 6| = 2|x - 3| < 2\delta = 2 \cdot \frac{\varepsilon}{2} = \varepsilon.$$

Since for every $\varepsilon > 0$ we found a valid $\delta > 0$ satisfying the definition, $\lim_{x \rightarrow 3} (2x - 1) = 5$. ■

Example 6.2: Proving a limit with a quadratic (bounded coefficient trick)

Prove $\lim_{x \rightarrow 2} x^2 = 4$.

Solution. *Scratch work:* $|x^2 - 4| = |x - 2||x + 2|$. We control $|x - 2|$ via δ , but $|x + 2|$ also varies with x . **Trick:** first restrict $\delta \leq 1$ so that x stays near 2, which bounds $|x + 2|$.

If $\delta \leq 1$ and $|x - 2| < \delta \leq 1$, then $1 < x < 3$, so $3 < x + 2 < 5$, hence $|x + 2| < 5$.

Now we want $|x - 2||x + 2| < \varepsilon$. Since $|x + 2| < 5$, it suffices to have $|x - 2| \cdot 5 < \varepsilon$, i.e. $|x - 2| < \frac{\varepsilon}{5}$.

Formal proof: Let $\varepsilon > 0$. Choose $\delta = \min\left(1, \frac{\varepsilon}{5}\right)$. Suppose $0 < |x - 2| < \delta$. Since $\delta \leq 1$, we have $|x + 2| < 5$ as shown above. Then

$$|x^2 - 4| = |x - 2||x + 2| < \delta \cdot 5 \leq \frac{\varepsilon}{5} \cdot 5 = \varepsilon.$$

Hence $\lim_{x \rightarrow 2} x^2 = 4$. ■

6.5 Continuity

6.5.1 The Big Idea

A function is **continuous** at a point if you could draw its graph through that point **without lifting your pen** – no holes, no jumps, no breaks.

Continuity at a Point

f is **continuous at** $x = a$ if all three of the following hold:

1. $f(a)$ is defined,
2. $\lim_{x \rightarrow a} f(x)$ exists,
3. $\lim_{x \rightarrow a} f(x) = f(a)$.

If f is continuous at every point of an interval, we say f is **continuous on that interval**.

ε - δ Definition of Continuity

f is continuous at a if for every $\varepsilon > 0$ there is a $\delta > 0$ such that

$$|x - a| < \delta \implies |f(x) - f(a)| < \varepsilon.$$

(Note: unlike the limit definition, we do **not** need $x \neq a$, since we want $f(x)$ close to $f(a)$, which is trivially true exactly at $x = a$.)

6.5.2 Types of Discontinuity

Type	What fails	Example
Removable	$\lim_{x \rightarrow a} f(x)$ exists but $\neq f(a)$, or $f(a)$ undefined	$f(x) = \frac{x^2-1}{x-1}$ at $x = 1$ (hole)
Jump	$\lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x)$	Piecewise step functions, $\lfloor x \rfloor$
Infinite	one-sided limit is $\pm\infty$	$f(x) = \frac{1}{x}$ at $x = 0$

Continuity Rules (built from Limit Laws)

If f and g are continuous at a , then so are: $f \pm g$, cf , fg , and f/g (provided $g(a) \neq 0$). Polynomials are continuous everywhere; rational functions are continuous everywhere except where the denominator is zero; $\sin x, \cos x, e^x$ are continuous everywhere; $\sqrt{x}$ is continuous on $[0, \infty)$; compositions of continuous functions are continuous.

6.6 Worked Examples – Continuity**Example 6.3: Checking continuity using the three-part test**

Is $f(x) = \begin{cases} \frac{x^2-4}{x-2}, & x \neq 2 \\ 5, & x = 2 \end{cases}$ continuous at $x = 2$?

Solution. (1) $f(2) = 5$, defined. ✓

(2) $\lim_{x \rightarrow 2} \frac{x^2-4}{x-2} = \lim_{x \rightarrow 2} (x+2) = 4$, exists. ✓

(3) Compare: $\lim_{x \rightarrow 2} f(x) = 4$, but $f(2) = 5$. Since $4 \neq 5$, condition (3) **fails**.

f is **not continuous** at $x = 2$ (a removable discontinuity – redefining $f(2) = 4$ would fix it).

Example 6.4: Finding a constant for continuity

Find k so that $f(x) = \begin{cases} kx + 1, & x \leq 2 \\ x^2 - 3, & x > 2 \end{cases}$ is continuous everywhere.

Solution. Both pieces are polynomials (continuous on their own domains), so the only possible discontinuity is at the joint $x = 2$. We need

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x) = f(2).$$

$$\lim_{x \rightarrow 2^-} (kx + 1) = 2k + 1, \quad \lim_{x \rightarrow 2^+} (x^2 - 3) = 4 - 3 = 1.$$

Set equal: $2k + 1 = 1 \implies k = 0$.

Check: with $k = 0$, $f(2) = 0(2) + 1 = 1$, matching both sides. So $k = 0$ makes f continuous everywhere.

6.7 Practice Problems – Section 6

Easy

1. Use the ε - δ definition to prove $\lim_{x \rightarrow 1} (3x + 2) = 5$.
2. Determine whether $f(x) = \frac{1}{x-1}$ is continuous at $x = 1$, and classify the discontinuity if any.
3. State the three conditions for continuity at a point.

Medium

4. Use ε - δ to prove $\lim_{x \rightarrow -1} (4 - 3x) = 7$.
5. Find k so that $g(x) = \begin{cases} x^2 + k, & x < 1 \\ 3x, & x \geq 1 \end{cases}$ is continuous at $x = 1$.
6. Classify the discontinuity of $h(x) = \frac{x+3}{x^2-9}$ at $x = 3$ and at $x = -3$.

Hard

7. Use ε - δ to prove $\lim_{x \rightarrow 1} (x^2 + x) = 2$ (use the bounded-coefficient trick from Example 6.2).
8. A function f satisfies $|f(x) - 3| \leq 2|x - 1|$ for all x near 1. Use the ε - δ definition to prove $\lim_{x \rightarrow 1} f(x) = 3$.

7 The Intermediate Value Theorem (IVT)

7.1 The Big Idea

If you walk continuously from the bottom of a hill (height 2m) to the top of a hill (height 100m), you must have passed through **every** height in between at some moment – say 50m – because you cannot teleport. This “no jumping” property of continuous functions is exactly what the Intermediate Value Theorem captures.

7.2 Formal Statement

Intermediate Value Theorem

Suppose f is **continuous on the closed interval** $[a, b]$, and let N be any number between $f(a)$ and $f(b)$ (with $f(a) \neq f(b)$). Then there exists at least one $c \in (a, b)$ such that

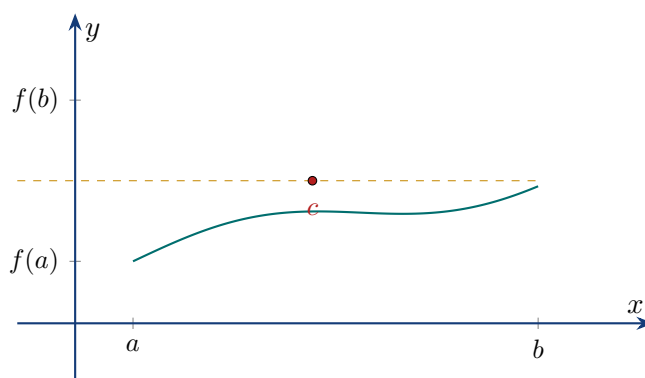
$$f(c) = N.$$

Key Note

Two conditions are essential:

- f must be **continuous** on the whole closed interval $[a, b]$ – IVT can fail if there’s even one break.
- N must lie **between** $f(a)$ and $f(b)$.

IVT guarantees **existence** of at least one c – it does *not* tell you the exact value of c , nor that c is unique (there could be several).



7.3 Key Application: Locating Roots

Root-Location Corollary

If f is continuous on $[a, b]$ and $f(a)$ and $f(b)$ have **opposite signs**, then f has at least one root (zero) $c \in (a, b)$, i.e. $f(c) = 0$. This follows from IVT with $N = 0$.

This is the theoretical basis for numerical root-finding methods (like the *bisection method*) which you will use extensively in Numerical Methods / programming courses.

7.4 Worked Examples

Example 7.1: Proving a root exists

Show that $f(x) = x^3 - 4x + 2$ has a root in the interval $(0, 1)$.

Solution. f is a polynomial, so it is continuous everywhere, in particular on $[0, 1]$.

$$f(0) = 0 - 0 + 2 = 2 > 0.$$

$$f(1) = 1 - 4 + 2 = -1 < 0.$$

Since $f(0) > 0$ and $f(1) < 0$ have opposite signs, and f is continuous on $[0, 1]$, by the IVT (with $N = 0$) there exists $c \in (0, 1)$ such that $f(c) = 0$. So f has a root in $(0, 1)$. ■

Example 7.2: Narrowing the interval (bisection idea)

Continuing Example 7.1, narrow the interval containing a root of $f(x) = x^3 - 4x + 2$ to width 0.25.

Solution. We know a root lies in $(0, 1)$. Test the midpoint $x = 0.5$:

$$f(0.5) = 0.125 - 2 + 2 = 0.125 > 0.$$

Since $f(0.5) > 0$ and $f(1) < 0$, the root lies in $(0.5, 1)$. Test midpoint $x = 0.75$:

$$f(0.75) = 0.421875 - 3 + 2 = -0.578125 < 0.$$

Since $f(0.5) > 0$ and $f(0.75) < 0$, the root lies in $(0.5, 0.75)$ – width 0.25, as required.

Example 7.3: Guaranteeing a specific intermediate value

Let $f(x) = x^2$ on $[1, 3]$. Show that f takes the value 5 somewhere in $(1, 3)$, and find that x -value exactly.

Solution. f is continuous (polynomial) on $[1, 3]$. $f(1) = 1$, $f(3) = 9$. Since $1 < 5 < 9$, by IVT there exists $c \in (1, 3)$ with $f(c) = 5$.

Here we can solve exactly: $c^2 = 5 \Rightarrow c = \sqrt{5} \approx 2.236$, which indeed lies in $(1, 3)$. ✓

Common Mistake

IVT **requires continuity on the entire closed interval**. If f has even one discontinuity inside $[a, b]$, the conclusion can fail – e.g. $f(x) = 1/x$ has $f(-1) = -1 < 0$ and $f(1) = 1 > 0$, opposite signs, yet f **never equals zero** anywhere, because f is discontinuous at $x = 0 \in (-1, 1)$.

7.5 Practice Problems – Section 7

Easy

1. Show that $f(x) = x^3 - x - 1$ has a root in $(1, 2)$.
2. Show $f(x) = \cos x - x$ has a root in $(0, 1)$ (use $f(0) = 1$, $f(1) = \cos 1 - 1 \approx -0.46$).
3. State the two hypotheses required to apply the IVT.

Medium

4. Show that $f(x) = x^5 - 2x^3 + 1$ takes the value 0 somewhere in $(-2, 0)$.

5. Show $f(x) = x^2 - 2$ takes the value 0 in $(1, 2)$, and hence give an IVT-based argument for the existence of $\sqrt{2}$.
6. Use IVT to show $x = \cos x$ has a solution in $(0, \pi/2)$.

Hard

7. Narrow a root of $f(x) = x^3 - x - 1$ (Q1 above) to an interval of width 0.25 using repeated bisection.
8. Let f be continuous on $[0, 1]$ with $f(0) = f(1)$. Prove that there exists $c \in [0, \frac{1}{2}]$ such that $f(c) = f(c + \frac{1}{2})$ (hint: consider $g(x) = f(x + \frac{1}{2}) - f(x)$ on $[0, \frac{1}{2}]$).

8 Limits at Infinity and Horizontal Asymptotes

8.1 The Big Idea

Instead of asking what happens as x approaches a finite point, we now ask: what happens to $f(x)$ as x grows without bound, either towards $+\infty$ or towards $-\infty$? This describes the **end behaviour** of a function – useful for understanding long-term trends (population growth, algorithm running time as input size $n \rightarrow \infty$, etc.).

8.2 Definition

Limits at Infinity

$\lim_{x \rightarrow \infty} f(x) = L$ means $f(x)$ gets arbitrarily close to L as x increases without bound. Similarly, $\lim_{x \rightarrow -\infty} f(x) = L$ means $f(x) \rightarrow L$ as x decreases without bound (towards $-\infty$).

Horizontal Asymptote

The line $y = L$ is a **horizontal asymptote** of $y = f(x)$ if

$$\lim_{x \rightarrow \infty} f(x) = L \quad \text{or} \quad \lim_{x \rightarrow -\infty} f(x) = L.$$

A function can have zero, one, or two (different) horizontal asymptotes.

8.3 Key Tool: Dividing by the Highest Power

Basic Fact and Main Technique

$$\lim_{x \rightarrow \pm\infty} \frac{1}{x^n} = 0 \quad \text{for any } n > 0.$$

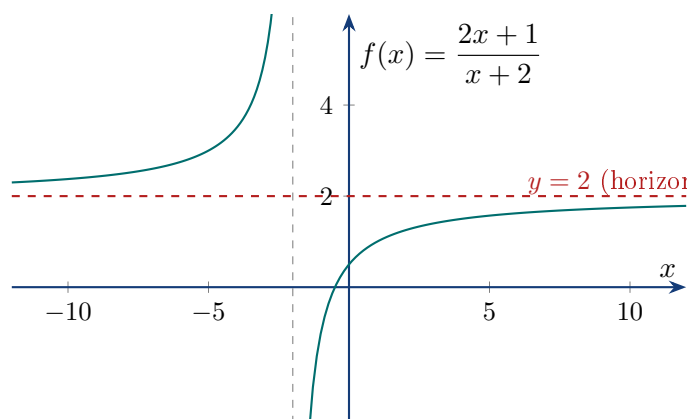
For rational functions $f(x) = \frac{p(x)}{q(x)}$, **divide every term by the highest power of x appearing in the denominator**, then apply the fact above term by term.

Shortcut Rules for Rational Functions – Comparing Degrees

Let $p(x)$ have degree m with leading coefficient A , and $q(x)$ have degree n with leading coefficient B , for $f(x) = p(x)/q(x)$.

- If $m < n$ (numerator degree smaller): $\lim_{x \rightarrow \pm\infty} f(x) = 0$. Horizontal asymptote $y = 0$.
- If $m = n$ (equal degrees): $\lim_{x \rightarrow \pm\infty} f(x) = \frac{A}{B}$ (ratio of leading coefficients). Horizontal asymptote $y = A/B$.
- If $m > n$ (numerator degree larger): the limit is $\pm\infty$ (no horizontal asymptote; check for a slant/oblique asymptote instead if $m = n + 1$).

8.4 Graphical Illustration



Here, numerator and denominator both have degree 1, so $\lim_{x \rightarrow \pm\infty} \frac{2x+1}{x+2} = \frac{2}{1} = 2$, giving horizontal asymptote $y = 2$ (matching both ends of the graph).

8.5 Worked Examples

Example 8.1: Equal degrees

Evaluate $\lim_{x \rightarrow \infty} \frac{3x^2 - 5x + 1}{2x^2 + 7}$.

Solution. Divide every term by x^2 (the highest power in the denominator):

$$\frac{3x^2 - 5x + 1}{2x^2 + 7} = \frac{3 - \frac{5}{x} + \frac{1}{x^2}}{2 + \frac{7}{x^2}}.$$

As $x \rightarrow \infty$, $\frac{5}{x} \rightarrow 0$, $\frac{1}{x^2} \rightarrow 0$, $\frac{7}{x^2} \rightarrow 0$:

$$\lim_{x \rightarrow \infty} \frac{3x^2 - 5x + 1}{2x^2 + 7} = \frac{3 - 0 + 0}{2 + 0} = \frac{3}{2}.$$

(Matches the shortcut: equal degrees, ratio of leading coefficients $3/2$.)

Example 8.2: Numerator degree smaller

Evaluate $\lim_{x \rightarrow -\infty} \frac{4x + 1}{x^2 - 3}$.

Solution. Divide by x^2 (highest power in denominator):

$$\frac{4x + 1}{x^2 - 3} = \frac{\frac{4}{x} + \frac{1}{x^2}}{1 - \frac{3}{x^2}}.$$

As $x \rightarrow -\infty$, every term with x in the denominator $\rightarrow 0$:

$$\lim_{x \rightarrow -\infty} \frac{4x + 1}{x^2 - 3} = \frac{0 + 0}{1 - 0} = 0.$$

Horizontal asymptote: $y = 0$.

Example 8.3: Square roots – careful with sign as x approaches negative infinity

Evaluate $\lim_{x \rightarrow -\infty} \frac{\sqrt{9x^2 + 1}}{x}$.

Solution. Careful: for $x < 0$, $\sqrt{x^2} = |x| = -x$ (not x !). Write $\sqrt{9x^2 + 1} = \sqrt{x^2} \sqrt{9 + \frac{1}{x^2}} = |x| \sqrt{9 + \frac{1}{x^2}}$.

Since $x \rightarrow -\infty$ means $x < 0$, we have $|x| = -x$:

$$\frac{\sqrt{9x^2 + 1}}{x} = \frac{-x \sqrt{9 + \frac{1}{x^2}}}{x} = -\sqrt{9 + \frac{1}{x^2}}.$$

As $x \rightarrow -\infty$, $\frac{1}{x^2} \rightarrow 0$:

$$\lim_{x \rightarrow -\infty} \frac{\sqrt{9x^2 + 1}}{x} = -\sqrt{9} = -3.$$

Example 8.4: Numerator degree larger

Evaluate $\lim_{x \rightarrow \infty} \frac{x^3 - 2x}{5x^2 + 1}$.

Solution. Numerator degree (3) > denominator degree (2), so we expect $\pm\infty$. Divide by x^2 :

$$\frac{x^3 - 2x}{5x^2 + 1} = \frac{x - \frac{2}{x}}{5 + \frac{1}{x^2}}.$$

As $x \rightarrow \infty$: numerator $\rightarrow \infty$ (since $x \rightarrow \infty$ dominates), denominator $\rightarrow 5$.

$$\lim_{x \rightarrow \infty} \frac{x^3 - 2x}{5x^2 + 1} = \infty.$$

No horizontal asymptote in this direction (the graph has a slant asymptote instead, a topic in later chapters).

Common Mistake

When a square root is involved, remember $\sqrt{x^2} = |x|$, **not** x . This sign subtlety is the single most common mistake in limits-at-infinity problems involving radicals – always split into $x \rightarrow +\infty$ ($|x| = x$) and $x \rightarrow -\infty$ ($|x| = -x$) cases separately.

8.6 Practice Problems – Section 8**Easy**

1. Evaluate $\lim_{x \rightarrow \infty} \frac{5x + 2}{3x - 1}$.

2. Evaluate $\lim_{x \rightarrow \infty} \frac{7}{x^2 + 1}$.

3. State the horizontal asymptote(s) of $f(x) = \frac{3x - 1}{x + 4}$.

Medium

4. Evaluate $\lim_{x \rightarrow \infty} \frac{2x^2 - 3}{5x^3 + x}$.
5. Evaluate $\lim_{x \rightarrow \infty} \frac{\sqrt{4x^2 + 3}}{2x - 1}$.
6. Evaluate $\lim_{x \rightarrow -\infty} (x^3 - 100x^2)$ by factoring out the dominant term.

Hard

7. Evaluate $\lim_{x \rightarrow \infty} \left(\sqrt{x^2 + 3x} - x \right)$ (hint: rationalize by multiplying by the conjugate).
8. Evaluate $\lim_{x \rightarrow -\infty} \frac{\sqrt{x^2 + 1}}{5x + 2}$, carefully tracking the sign of x .

9 Important Questions for the CSE 1-1 Semester Exam

Based on the typical emphasis of a first-year Bangladeshi engineering-university calculus course, the following question types are **high-priority** for exam preparation. Practice each type until you can do it without hesitation.

1. **Evaluate limits by factoring / rationalizing** ($0/0$ indeterminate forms) – appears almost every exam (see Section 4).
2. **ε - δ proof of a given limit**, usually linear or simple quadratic (Section 6, Examples 6.1–6.2) – a guaranteed theory question.
3. **Find k (or another constant) so a piecewise function is continuous** at the joint point (Section 6, Example 6.4) – extremely common.
4. **State and apply the Sandwich Theorem** to a function involving $\sin(1/x)$ or $\cos(1/x)$ (Section 5) – classic combined theory+numerical question.
5. **State the IVT and use it to show a root exists** in a given interval, often for a cubic (Section 7, Example 7.1) – guaranteed application question.
6. **Limits at infinity of rational functions**, including with square roots – know the degree-comparison shortcut cold (Section 8).
7. **One-sided limits of piecewise/floor/absolute-value functions**, and explaining why the two-sided limit does/doesn't exist (Section 2).
8. **Infinite limits and identifying vertical asymptotes** of rational functions via sign analysis (Section 3).
9. **Definitions – write from memory**: formal ε - δ limit definition, definition of continuity at a point, statement of IVT, statement of Sandwich Theorem. These are commonly worth direct marks.
10. **Short conceptual/true-false style questions**, e.g.: “Can a limit exist if the function is undefined at that point?”, “Does $f(a)$ being defined guarantee continuity?”, “Is ∞ a real number?”

Key Note

Exam strategy tip: Always try direct substitution *first*. If you get a real number, you are done. If you get $\frac{0}{0}$, factor or rationalize. If you get a nonzero number over zero, it is an infinite limit – do sign analysis. If $x \rightarrow \pm\infty$ appears, divide by the highest power in the denominator.

10 Answer Key

Brief final answers / key solution steps only. Refer back to the corresponding section's worked examples for full method if you get stuck.

Section 1 – Basic Concepts of Limits

1. $\lim_{x \rightarrow 3} (2x + 1) = 7$.
2. Factor $\Rightarrow x + 1 \Rightarrow \text{limit} = 2$.
3. Yes – e.g. $f(x) = \frac{x^2-1}{x-1}$ at $x = 1$: $f(1)$ undefined, but limit = 2. The limit only depends on values near a .
4. Factor: $\frac{(x-2)(x^2+2x+4)}{x-2} = x^2 + 2x + 4 \rightarrow 12$.
5. Any function with a removable discontinuity or redefinition at $x = 2$, e.g. $f(x) = x + 1$ for $x \neq 2$, $f(2) = 5$.
6. Rationalize: multiply by $\frac{\sqrt{x+2}}{\sqrt{x+2}}$; simplifies to $\frac{1}{\sqrt{x+2}} \rightarrow \frac{1}{4}$.
7. $\sin(1/x)$ oscillates between -1 and 1 infinitely often near 0 ; multiplying by $1/x$ makes the oscillation amplitude blow up (unbounded), so it never settles – DNE.
8. E.g. $g(x) = 2$ for irrational x near 0 and g undefined for rational x near 0 except $x = 0$ where $g(0) = 2$; informally, any function equal to 2 on a "thinning" set around 0 approaching fully.

Section 2 – One-Sided Limits

1. $\lim_{x \rightarrow 0^-} f = 2$, $\lim_{x \rightarrow 0^+} f = 0$.
2. No – $\sqrt{x}$ is undefined for $x < 0$, so only $\lim_{x \rightarrow 0^+}$ makes sense; the two-sided limit is undefined/DNE by convention since the left side isn't in the domain.
3. Both equal 5 (constant floor value near non-integer point) – actually $\lim_{x \rightarrow 5^-} \lfloor x \rfloor = 4$, $\lim_{x \rightarrow 5^+} \lfloor x \rfloor = 5$.
4. Limit as $x \rightarrow 2$ of $2x + 1$ is 5 and of $x^2 - 3$ is 1 ; they disagree, so **no** value of k makes the two-sided limit exist (only g defined at 2 can be set to anything, doesn't fix the DNE two-sided limit).
5. $\lim_{x \rightarrow 2^-} = -1$, $\lim_{x \rightarrow 2^+} = 1$.
6. $\lim_{x \rightarrow 3^-} h = 6$, $\lim_{x \rightarrow 3^+} h = 6$, so two-sided limit is $6 \neq h(3) = 10$: removable discontinuity, limit $\neq h(3)$.
7. Example: $f(x) = \frac{x^2-1}{x-1}$ for $x \neq 1, 3$; $f(1) = 10$ (removable at 1); $f(x) = x$ for $x < 3$, $f(x) = x + 2$ for $x \geq 3$ (jump at 3). Compute limits accordingly.
8. $\lfloor x \rfloor + \lfloor -x \rfloor = 0$ for integer n but $= -1$ for non-integer x ; so $\lim_{x \rightarrow n^-} f = -1$, $\lim_{x \rightarrow n^+} f = -1$, both differ from $f(n) = 0$ – removable-type jump at every integer.

Section 3 – Infinite Limits

1. $\lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty$, $\lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty$.
2. Vertical asymptote at $x = -4$.
3. $\lim_{x \rightarrow 0} \frac{4}{x^2} = +\infty$ (both sides, since $x^2 > 0$).
4. Denominator $= (x - 3)(x + 2)$. At $x = 3$: left $\rightarrow -\infty$, right $\rightarrow +\infty$. At $x = -2$: left $\rightarrow +\infty$, right $\rightarrow -\infty$.
5. $\lim_{x \rightarrow 2^+} \frac{x}{x-2} = +\infty$, $\lim_{x \rightarrow 2^-} \frac{x}{x-2} = -\infty$.
6. No vertical asymptote – numerator also zero at $x = 3$ (removable hole); simplifies to $x + 3 \rightarrow 6$.
7. Factor denominator: $x(x - 4)(x + 1)$; numerator $(x - 2)(x + 2)$. Vertical asymptotes at $x = 0, 4, -1$ (none cancel). Sign-check each side individually using the factored sign chart.

- Two vertical asymptotes at $x = a$ and $x = b$ (assuming $a \neq b$); near each, sign flips from $-\infty$ to $+\infty$ or vice versa depending on the sign of the other term at that point.

Section 4 – Limit Laws

- $3(4) - 10 + 1 = 3$.
- Factor denominator: $\frac{x+3}{(x-3)(x+3)} = \frac{1}{x-3} \rightarrow \frac{1}{-6} = -\frac{1}{6}$.
- $3(5) - 2(2) = 11$.
- $\frac{1}{4}$ (same as Example 4.2 method).
- Expand $(2+x)^3 - 8 = 12x + 6x^2 + x^3$; divide by x : $12 + 6x + x^2 \rightarrow 12$.
- Expand: $(3+h)^2 - 9 = 6h + h^2$; divide by h : $6 + h \rightarrow 6$.
- Rationalize denominator: multiply by $\frac{\sqrt{x+1}}{\sqrt{x+1}}$, numerator factors as $(x-1)(x^2+x+1)$, denominator becomes $x-1$; simplifies to $(\sqrt{x+1})(x^2+x+1) \rightarrow 2 \cdot 3 = 6$.
- Combine over common denominator, rationalize; result is $-\frac{1}{2}$.

Section 5 – Sandwich Theorem

- 0 (squeeze between $\pm x^2$).
- 0 (squeeze between $\pm x$).
- 2 (squeeze between $2 - x^2 \rightarrow 2$ and $2 + x^2 \rightarrow 2$).
- 0 (squeeze between 0 and x^2 , using $\sin^2 \leq 1$).
- 1 (squeeze between $1 - x^2/6 \rightarrow 1$ and $1 \rightarrow 1$).
- 0 (squeeze between $-1/x \rightarrow 0$ and $1/x \rightarrow 0$ as $x \rightarrow \infty$).
- 0 (squeeze between $-x^2e^{-1}$ and x^2e , both $\rightarrow 0$... more precisely between $-x^2e$ and x^2e , both $\rightarrow 0$).
- Divide inequality by $\theta > 0$: $\cos \theta \leq \frac{\sin \theta}{\theta} \leq 1$; as $\theta \rightarrow 0^+$, $\cos \theta \rightarrow 1$, squeeze gives limit = 1.

Section 6 – Formal Definitions & Continuity

- Choose $\delta = \varepsilon/3$; standard linear proof.
- Not continuous at $x = 1$ – infinite discontinuity (vertical asymptote), $f(1)$ undefined.
- $f(a)$ defined; $\lim_{x \rightarrow a} f(x)$ exists; the two are equal.
- Choose $\delta = \varepsilon/3$ (since $|-3x + 3| = 3|x + 1|$).
- Need $1 + k = 3 \Rightarrow k = 2$.
- At $x = 3$: numerator nonzero, denominator zero $\Rightarrow$ infinite discontinuity. At $x = -3$: both zero $\Rightarrow$ removable discontinuity.
- Choose $\delta = \min(1, \varepsilon/4)$ using bounded-coefficient trick (bound $|x + 2| < 4$ near $x = 1$).
- Given $\varepsilon > 0$, choose $\delta = \varepsilon/2$; then $|f(x) - 3| \leq 2|x - 1| < 2\delta = \varepsilon$.

Section 7 – Intermediate Value Theorem

- $f(1) = -1 < 0$, $f(2) = 5 > 0 \Rightarrow$ root in $(1, 2)$ by IVT.
- $f(0) = 1 > 0$, $f(1) \approx -0.46 < 0 \Rightarrow$ root in $(0, 1)$.
- (i) f continuous on $[a, b]$; (ii) N between $f(a)$ and $f(b)$.
- $f(-2) = -32 + 16 + 1 = -15 < 0$, $f(0) = 1 > 0 \Rightarrow$ root in $(-2, 0)$.
- $f(1) = -1 < 0$, $f(2) = 2 > 0 \Rightarrow$ root in $(1, 2)$, i.e. $\sqrt{2}$ exists as this root.
- $g(x) = \cos x - x$: $g(0) = 1 > 0$, $g(\pi/2) = -\pi/2 < 0 \Rightarrow$ root in $(0, \pi/2)$.
- Bisection: test $x = 1.5$: $f(1.5) = 0.875 > 0$; root in $(1, 1.5)$; test $x = 1.25$: $f(1.25) \approx -0.30 < 0$; root in $(1.25, 1.5)$ – width 0.25.

8. Apply IVT to $g(x) = f(x + \frac{1}{2}) - f(x)$ on $[0, \frac{1}{2}]$: $g(0) = f(\frac{1}{2}) - f(0)$ and $g(\frac{1}{2}) = f(1) - f(\frac{1}{2}) = f(0) - f(\frac{1}{2}) = -g(0)$; since $g(0)$ and $g(1/2)$ are negatives of each other, either both zero or opposite signs, so by IVT $g(c) = 0$ for some $c \in [0, \frac{1}{2}]$.

Section 8 – Limits at Infinity

1. Equal degrees: $5/3$.
2. Numerator degree smaller: 0 .
3. $y = 3$ (equal degrees, ratio $3/1$).
4. Numerator degree smaller: 0 .
5. Divide by x (careful with $\sqrt{}$): as $x \rightarrow \infty$, $\rightarrow \frac{2}{2} = 1$.
6. Factor x^3 : $x^3(1 - 100/x) \rightarrow -\infty$ (since $x^3 \rightarrow -\infty$ dominates).
7. Rationalize: $\frac{3x}{\sqrt{x^2+3x+x}}$; divide by x : $\rightarrow \frac{3}{1+1} = \frac{3}{2}$.
8. For $x \rightarrow -\infty$, $\sqrt{x^2+1} = |x|\sqrt{1+1/x^2} = -x\sqrt{1+1/x^2}$; expression $\rightarrow \frac{-x}{5x} = -\frac{1}{5}$.